👩‍🏫 Staff Guide · Small Engines Overhaul · Unit 3 4-Stroke Theory & Engine Parts (Level 1)

Level 1 = about 2nd-grade reading, with emoji and pictures. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Week 2: 4 Stroke Theory PowerPoint (Small Engines Parts ID Reading) + Parts ID quiz, Engine Tools/Parts sheets, Briggs & Stratton disassembly/assembly video, Marine 2-stroke and 4-stroke video notes, iCADEMY Bronze 6 Engine Principles of Operation")	**Level 1 piece**
4 Stroke Theory PowerPoint (13 slides) + Parts ID quiz	Deck + notes A, B + quiz; Parts ID lab packet
Engine Tools/Parts sheets	Parts ID lab packet + vocab cards
B&S disassembly/assembly video	Video guide (link only)
Marine 2-stroke/4-stroke video notes	Slides "2-Stroke vs 4-Stroke" + notes C
iCADEMY Bronze 6 Engine Principles	Our own slides on the 4 strokes (no STIHL content)
Class slides	Deck: 14 slides, Words to Know first, "point to the picture" checks, recap
His worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 8 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- **Read aloud** every line; student may point, say, or circle (point = full answer)
- One idea per slide; big pictures; 2–3 picture choices
- Act out the 4 strokes: students move hands (suck, squeeze, bang, blow)
- Cutaway engine or clear model if the shop has one

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part
- Cover one choice to make it 2

4. What to grade

- Parts ID quiz: L1 points to 8 parts on a real engine or picture
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say "check with the class teacher".

🚧 Under construction (check with the class teacher)

- **iCADEMY Bronze 6**: STIHL iCADEMY is login-only; we wrote our own engine-principles content.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Engine Parts ID (Level 1)

Small Engines Overhaul · Unit 3 · goes with the class “Small Engine Parts ID Quiz + Engine Tools/Parts sheets” (quiz points). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Small Engine Parts ID Quiz + Engine Tools/Parts sheets”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Model first, then the student tries with adult supervision
- Picture task card at the bench
- Parts tray with labeled cups for small parts

4. What to grade (quiz points)

- Did the steps with support (photo proof)
- My Check page done
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
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⚙️ 4-Stroke Theory & Engine Parts Notes

Small Engines Overhaul · Unit 3 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

⬇️ intake _____

💥 power _____

🔩 piston _____

🚪 valve _____

🖼️ B. Fill in the word. Use the pictures.

5. ⬇️ Air and gas come in on _____.

6. 💥 The spark gives _____.

7. 🔩 The _____ goes up and down.

8. 🚪 A _____ opens and closes.

Word bank: ⬇️ **intake** | 🔩 **piston** | 💥 **power** | 🚪 **valve**

⭕ C. Circle the right picture.

9. Smoke goes out on…

⬇️ Intake	💨 Exhaust

10. The crankshaft…

🧊 Freezes	🔄 Spins

11. A chainsaw is a…

2️⃣ 2-stroke	🚲 Bicycle

🔗 D. Draw a line from the picture to the word.

⬇️	compression
🗜️	power
💥	exhaust
💨	intake

📖 4-Stroke Theory & Engine Parts: Vocabulary Picture Cards (Level 1)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Intake

air and gas come in



Compression

squeeze the mix



Power

spark makes it push



Exhaust

smoke goes out



Piston

moves up and down



Crankshaft

turns up-down into spinning



Valve

a door for gas



Flywheel

heavy wheel that keeps it turning

🔩 Engine Parts ID

Small Engines Overhaul · Unit 3 · Level 1



Name: _____

Modified from the class “Small Engine Parts ID Quiz + Engine Tools/Parts sheets”.

🧺 Supplies: get it. Check ☐.

- ☐ ⚙️ **Practice engine** staff choose
- ☐ 🏷️ **Part label cards**
- ☐ 🧰 **Tool tray**
- ☐ 📋 **Parts ID sheet**
- ☐ 📸 **Camera/iPad**

🔩 Do one box. Check ☐ it.

1		Glasses on.	☐
2		Adult checks plug wire is off.	☐
3		Look at the engine.	☐
4		Put a card on the spark plug.	☐
5		Put a card on the air filter.	☐
6		Card on flywheel, muffler, gas tank.	☐
7		Take a photo.	☐

🔩 Engine Parts ID · Get Ready

Small Engines Overhaul · Unit 3 · Level 1



Name: _____

👥 My job. Circle it.

- **Mechanic** — fixes
- **Tool helper** — gets tools
- **Recorder** — writes
- **Clean-up captain** — cleans

🧰 Get the tools. Check ☐.

- ☐ Practice engine
- ☐ Part label cards
- ☐ Tool tray
- ☐ Parts ID sheet
- ☐ Camera/iPad

⚠️ Be safe!

- 🥽 Glasses on.
- 👖 Coveralls on.
- 🔧 Tools with an adult.
- 🔌 Plug wire off.
- 🧼 Clean up, wash hands.

✅ Before we start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

🔩 Engine Parts ID · My Check

Small Engines Overhaul · Unit 3 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 📸 Before
- ☐ 📸 During
- ☐ 📸 After

✅ I did it! Check ☐.

- ☐ 🥽 I wore my safety gear
- ☐ 🔧 I did the engine job
- ☐ 🧽 I cleaned up
- ☐ 🤝 I helped my group

⭕ Circle one face.

🥽 **I was safe.**	😀	🙂	😕
🔧 **I did my job.**	😀	🙂	😕
⭐ **I liked it.**	😀	🙂	😕

⭐ I learned ____. Draw or write.

✅ 4-Stroke Theory & Engine Parts Quiz

Small Engines Overhaul · Unit 3 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. ⬇️ Air and gas come in…

- 💥 Power
- ⬇️ Intake

2. 🗜️ Squeeze is…

- 🗜️ Compression
- 💨 Exhaust

3. 💥 The spark makes…

- 💥 Power
- 🧊 Ice

4. 💨 Smoke goes out…

- ⬇️ Intake
- 💨 Exhaust

5. 🔩 Goes up and down…

- 🛞 Wheel on a car
- 🔩 Piston

6. 🔁 Spins…

- 🔁 Crankshaft
- 🚪 Door

7. ⚡ Makes the spark…

- 🌬️ Air filter
- ⚡ Spark plug

8. 🪚 Oil mixed in gas…

- 2️⃣ 2-stroke
- 4️⃣ 4-stroke

🎬 Engine Disassembly / Assembly Video

Small Engines Overhaul · Unit 3 · Level 1 video guide · class points pts

Name: _____

👀 Watch for these pictures.



Briggs & Stratton video linked in Canvas (YouTube). Link only.

1. First they take off the…

-  **Outside parts**
-  **Pizza**

2. They use…

-  **Wrenches**
-  **Crayons**

3. The heavy wheel is the…

-  **Balloon**
-  **Flywheel**

4. Inside is the…

-  **Toy**
-  **Piston**

⭐ **My favorite part:** _____

⚙️ 4-Stroke Theory & Engine Parts Notes · KEY

Small Engines Overhaul · Unit 3 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

⬇️ intake _____

💥 power _____

🔩 piston _____

🚪 valve _____

🖼️ B. Fill in the word. Use the pictures.

5. ⬇️ Air and gas come in on **intake**.

6. 💥 The spark gives **power**.

7. 🔩 The **piston** goes up and down.

8. 🚪 A **valve** opens and closes.

Word bank: ⬇️ **intake** | 🔩 **piston** | 💥 **power** | 🚪 **valve**

⭕ C. Circle the right picture.

9. Smoke goes out on…

- ⬇️ Intake
- 💨 Exhaust **(selected)**

10. The crankshaft…

- 🧊 Freezes
- 🔄 Spins **(selected)**

11. A chainsaw is a…

- 2️⃣ 2-stroke **(selected)**
- 🚲 Bicycle

🔗 D. Draw a line from the picture to the word.

Key: ⬇️ → intake · 🗜️ → compression · 💥 → power · 💨 → exhaust

⬇️	compression
🗜️	power
💥	exhaust
💨	intake

🔑 *Key written by our team from the class teacher's materials, please check.*

✅ 4-Stroke Theory & Engine Parts Quiz · KEY

Small Engines Overhaul · Unit 3 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. ⬇️ Air and gas come in… Parts ID quiz

- Power
- ⬇️ **Intake** ✓

2. 🗜️ Squeeze is… Parts ID quiz

- 🗜️ **Compression** ✓
- Exhaust

3. 💥 The spark makes… Parts ID quiz

- 💥 **Power** ✓
- Ice

4. 💨 Smoke goes out… Parts ID quiz

- Intake
- 💨 **Exhaust** ✓

5. 🔩 Goes up and down… Parts ID quiz

- Wheel on a car
- 🔩 **Piston** ✓

6. 🔁 Spins… Parts ID quiz

- [x] 🔁 Crankshaft
- [ ] 🚪 Door

7. ⚡ Makes the spark… Parts ID quiz

- [ ] 🌬️ Air filter
- [x] ⚡ Spark plug

8. 🪚 Oil mixed in gas… Marine video

- [x] 2️⃣ 2-stroke
- [ ] 4️⃣ 4-stroke

Answers

1-B (Intake) · 2-A (Compression) · 3-A (Power) · 4-B (Exhaust) · 5-B (Piston) · 6-A (Crankshaft) · 7-B (Spark plug) · 8-A (2-stroke)

Level 1 version of the gen ed “Small Engine Parts ID Quiz”. Read every question aloud. Score: __ / 8.

🔑 *Key written by our team from the class teacher's materials, please check.*

🎬 Engine Disassembly / Assembly Video · KEY

Small Engines Overhaul · Unit 3 · Level 1 video guide · class points pts

Name: _____

👀 Watch for these pictures.

🔧 🔩 ⚙️ 🔁

Briggs & Stratton video linked in Canvas (YouTube). Link only.

1. First they take off the…

- [x] 🔧 **Outside parts**
- [ ] 🍕 **Pizza**

2. They use…

- [x] 🔧 **Wrenches**
- [ ] 🖍️ **Crayons**

3. The heavy wheel is the…

- [ ] 🎈 **Balloon**
- [x] ⚙️ **Flywheel**

4. Inside is the…

- [ ] 🧸 **Toy**
- [x] 🔩 **Piston**

⭐ **My favorite part:** _____

Goes with the class “Engine Disassembly/Assembly Video” (class points pts). Same points when complete.
🔑 Key written by our team from the class teacher's materials, please check.